

Tymbolica

Exact symbolic computation inside Typst

User manual · version 0.1.0

Parse Typst mathematics, transform it with Symbolica's exact algebra engine,
and place the result directly back into a document.

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1 Start here

Tymbolica keeps symbolic calculation beside the mathematics it belongs to. You can write a formula in Typst, factor or differentiate it, and place the answer straight back into the page—without copying expressions to another program.

Here is the whole pattern: read some mathematics with `math`, do the algebra, and display the result with `to-typst`.

```
#import "@local/tymbolica:0.1.0": *

#let p = math($(x + y)^3 - (x^3 + y^3)$)
#to-typst(factor(expand(p)))
```

```
3

x

y

(x + y)
```

Tymbolica finds $3xy(x + y)$ exactly. The same three-step pattern—read, calculate, display—runs through the rest of this manual.

1.1 Installation

Until Tymbolica is published in Typst Universe, install it from a checkout. On Linux, place or symlink the repository root at:

```
~/local/share/typst/packages/local/tymbolica/0.1.0
```

Use the corresponding Typst data directory on macOS or Windows. The package root must contain `typst.toml`; its `typst` directory contains `lib.typ` and the bundled compressed engine and its loader. Then import:

```
#import "@local/tymbolica:0.1.0": *
```

From a source checkout, a document can instead import the library by relative path:

```
#import "path/to/tymbolica/typst/lib.typ": *
```

No build step is needed to use the package: everything required is already included in the checkout.

Why @local? The examples use `@local` because this version is installed from the repository. A future Typst Universe release will use its published namespace instead.

1.2 Create an engine

Tymbolica's Symbolica engine provides algebra, solving, evaluation, and matrices. It is stored as a compressed asset and expanded transparently by a small loader. Most operations are available directly from the imported top-level API. Create an engine with `init()` when you need a custom symbol namespace or parser grammar:

```
#let sym = init()
#let parse = sym.math
#let symbol = sym.symbol
#let x = symbol("x")
#let result = sym.factor(parse($x^2 - 1$))
```

Symbolic integration and its rule steps live in the companion `tymbolica-rubi` package, whose manual starts from Tymbolica Atom payloads.

1.3 Where to begin

If you want to...	Start with...
Turn Typst mathematics into a symbolic expression	math, symbol, and atom
Put an answer back into the document	to-typst
Factor, expand, differentiate, or take a series	expand, factor, derivative, series
Integrate and inspect Rubi's rule path	Use the companion tymbolica-rubi package
Replace a recurring symbolic pattern	wild, rule, replace
Evaluate a formula at many points	evaluate-many, evaluate-grid
Solve equations exactly or numerically	solve, nsolve, nsolve-system
Work with exact matrices	matrix, matrix-solve, row-reduce

2 A few ideas before the examples

2.1 Keep expressions symbolic until you display them

The value returned by `math` is a symbolic expression, not something Typst can typeset on its own. Pass it through as many algebraic operations as you like, then call `to-typst` where you want the answer to appear.

Typst `math` → `math` → symbolic expression → transform → `to-typst` → document

```
#let expression = math$(x + 1)^4$
#let transformed = expand(expression)
```

```
original: $ #to-typst(expression) $ \
expanded: $ #to-typst(transformed) $
```

original:

$$(1 + x)^4$$

expanded:

$$1 + 4x + 6x^2 + 4x^3 + x^4$$

For ordinary documents, `to-typst` is usually all you need. `to-latex` is handy when exporting to another system, while `canonical` shows Symbolica's plain text form when you are diagnosing a difficult expression.

2.2 Symbols and wildcards have different jobs

Function	Typical input	Use
<code>math</code>	<code>\$x^2 + 1\$</code>	Read a Typst formula.
<code>atom</code>	a number, string, formula, or expression	Turn a general value into an expression.
<code>symbol</code>	<code>"x"</code>	Create an ordinary symbol that belongs to the mathematics.
<code>function</code>	<code>"f"</code>	Create a callable function whose complete calls carry meta-data.
<code>wild</code>	<code>"a"</code>	Create a placeholder used only while matching a pattern.

The distinction that matters A variable is part of the formula. A wildcard is a blank in a pattern. Use `symbol("x")` for the x in $x^2 + 1$; use `wild("a")` when a replacement rule should accept any expression in place of a .

Repeated occurrences of the same wildcard in one pattern must capture the same expression. Different matches can bind it differently; the rewriting guide below turns that rule into a concrete example.

`symbol` can also carry information that should not alter the printed formula. Interpolate it with `#` when that information matters to the calculation:

```
#let mass = symbol("m", namespace: "model", tags: ("model::positive",))
#let shell = math($p^2 - #mass^2$)

$ p^2 - m^2 = #to-tyrst(shell) $
```

$$p^2 - m^2 = -m^2 + p^2$$

The page still shows an ordinary m . Behind it, a versioned envelope carries the exact Symbolica Atom plus an inspectable description containing its namespace and tags. The Atom is authoritative when `math` reads the formula; packages layered on top can give the tags their own meaning. Namespaced tags are registered on the Symbolica symbol itself, so they survive algebraic and cross-plugin round trips. There is no separate `notation.symbol` or `var` alias.

A content value is not callable in Typst, so a custom function head uses the separate function constructor. Its metadata belongs to the whole call:

```
#let response = function("R", namespace: "model", tags: ("model::response",))
#let time = symbol("t", namespace: "model")
#let value = math($#response(time) + 1$)

$ R(t) + 1 = #to-tyrst(value) $
```

$$R(t) + 1 = 1 + R(t)$$

The same identities can control document-side notation. Exact heads style a symbol everywhere; exact calls receive the complete function arguments and can choose a different layout. This changes only presentation—the returned content still carries the exact Atom.

```
#let x = symbol("x", namespace: "notation_example")
#let f = function("f", namespace: "notation_example")
#let expression = f(add(x, 1))
#let display = notation(
  heads: ("notation_example::x": $xi$),
  calls: {
    "notation_example::f": ctx => {
      let (argument,) = ctx.visual-arguments
      $cal(F)[#argument]$
    },
  },
)

$ #to-tyrst(expression, notation: display) $
```

$$\mathcal{F}[1 + \xi]$$

2.3 Built-in functions such as sine and cosine

Polynomials work with the imported top-level functions. For analytic functions such as `sin`, `cos`, and `exp`, create a Symbolica-flavoured set of functions so that differentiation and numerical evaluation recognize them:

```
#let sym = init(namespace: "symbolica")
#let parse = sym.math
#let symbol = sym.symbol
#let derivative = sym.derivative
#let render = sym.to-tyrst
#let x = symbol("x")

#render(derivative(parse($sin(x)$), x))
```

The worked pendulum example binds these functions to short local names so that the calculation remains readable.

2.4 Write equations as expressions equal to zero

Solver inputs are expressions understood to equal zero. For example, `math($x + y - 3$)` represents $x + y = 3$.

For several variables, each solution row follows the variable order you give the solver. Exact solutions are in the result's `branches` array. With symbolic parameters, check coverage: a "generic" result applies only where every expression in coverage-guard is nonzero. Numerical solving looks for one branch near your initial guess.

3 Worked mathematical guides

The quickest way to learn the package is to follow a calculation from question to answer. Each example below ends with a check, because a plausible-looking formula is not yet a convincing result.

3.1 Calibrate a pendulum model

A pendulum gives us a small but complete modelling problem. Begin with the potential $V(\theta) = \kappa(1 - \cos \theta)$, derive the restoring torque, and take a cubic small-angle approximation. Two torque readings are then enough to recover both the unknown scale κ and a sensor offset τ_0 .

The calculation follows the [pendulum-calibration example in Symbolica's repository README](#). We create the `symbolica` set of functions because `cos` must be understood as the analytic cosine rather than as an arbitrary function name.

3.1.1 Derive the model

```
#let sym = init(namespace: "symbolica")
#let (
  math: m, symbol: v, to-typtst: render,
  derivative, series, neg, add,
) = sym

#let potential = m($kappa (1 - cos(theta)))$
#let q = v("θ")
#let k = v("κ")
#let b = v("τ₀")

#let torque = neg(derivative(potential, q))
#let small-angle = series(torque, q, 0, 3)
#let model = add(small-angle, b)

$ V(theta) = #render(potential) $ \
$ tau(theta) = -(partial V)/(partial theta) = #render(torque) $ \
$ tau_small(theta) + tau_0 = #render(model) $
```

$$V(\theta) = \kappa(1 - \cos(\theta))$$

$$\tau(\theta) = -\frac{\partial V}{\partial \theta} = -1 \kappa \sin(\theta)$$

$$\tau_{\text{small}(\theta)} + \tau_0 = -\kappa \theta + \frac{1}{6} \kappa \theta^3 + \tau_0$$

The derivative gives the restoring torque, while the cubic series keeps the first nonlinear correction to the familiar small-angle law. Adding τ_0 leaves a model that is linear in the two unknown parameters.

3.1.2 Fit two readings exactly

```
#let q1 = v("θ₁")
#let q2 = v("θ₂")
#let t1 = v("τ₁")
#let t2 = v("τ₂")

#let fit = solve((
  sub(replace(model, q, q1), t1),
  sub(replace(model, q, q2), t2),
), (k, b)).branches.first().values

$ kappa = #render(fit.at(0)) $ \
$ tau_0 = #render(fit.at(1)) $
```

$$\kappa = (6\tau_1 - 6\tau_2)(-6\theta_1 + 6\theta_2 + \theta_1^3 - \theta_2^3)^{-1}$$

$$\tau_0 = (-6\theta_1 + 6\theta_2 + \theta_1^3 - \theta_2^3)^{-1}(-6\theta_1\tau_2 + 6\theta_2\tau_1 + \theta_1^3\tau_2 - \theta_2^3\tau_1)$$

Nothing has been rounded: both fitted parameters are still formulas in the angles and measured torques. Now insert two observations, $(\theta_1, \tau_1) = (0.10, -0.4697)$ and $(\theta_2, \tau_2) = (0.20, -0.9545)$.

3.1.3 Insert the measurements and check the fit

```
#let fitted = evaluate-many(
  fit,
  (q1, q2, t1, t2),
  ((0.10, 0.20, -0.4697, -0.9545)),
).first()
#let predictions = evaluate-many(
  model,
  (q, k, b),
  (
    (0.10, fitted.at(0).re, fitted.at(1).re),
    (0.20, fitted.at(0).re, fitted.at(1).re),
  ),
)

$ kappa approx #calc.round(fitted.at(0).re, digits: 6), quad
tau_0 approx #calc.round(fitted.at(1).re, digits: 6) $
predicted torques:
#predictions.map(row => str(calc.round(row.first().re, digits: 4))).join(", ")
```

$$\kappa \approx 4.905228, \quad \tau_0 \approx 0.020005$$

predicted torques: -0.4697, -0.9545

The formula for κ remains exact until the measured values are inserted. Substituting the fitted parameters back into the model reproduces both torque readings, so the final line checks the entire chain from potential to fit.

3.2 Rewrite a repeated identity with wildcards

This is where a wildcard earns its keep. Suppose an expression contains several copies of $\sin^2(a) + \cos^2(a)$, but the argument a is different each time. The pattern should insist that the two functions share an argument without fixing what that argument is.

```
#let source = math(
  $3 (sin(x)^2 + cos(x)^2)
  + sin(x + y)^2 + cos(x + y)^2$
)
#let identity = math(
  $sin("a_")^2 + cos("a_")^2$
)

#let one-pass = replace(source, identity, 1)
#let reduced = replace(source, identity, 1, repeat: true)

source: #to-typst(source)\
one pass: #to-typst(one-pass)\
repeat to a fixed point: #to-typst(reduced)
```

source: 3

$(\sin(x)^2 + \cos(x)^2) + \sin(x + y)^2 + \cos(x + y)^2$
one pass: 1+ 3

$(\sin(x)^2 + \cos(x)^2)$
repeat to a fixed point: 4

Within one match, both occurrences of `a_` capture the same expression. Across matches it first captures one argument and then another. This is the practical difference between an ordinary symbol (`"a"`) and `wild("a")`.

Repeating rules `repeat: true` keeps applying the rule until the expression stops changing. Use it only for rules that settle: $a \rightarrow a + 1$, for example, would never reach a fixed point.

3.3 Separate a rational response into modes

Suppose a response function arrives in a form with one removable factor and two remaining poles. Cancel the shared factor, split the reduced response into simple modes, and then recombine the modes to check that nothing was lost. This follows the [rational-expression workflow in Symbolica's First Steps](#).

```
#let s = symbol("s")
#let response = math((s + 3)(2 s + 5)) / (s^3 + 6 s^2 + 11 s + 6)$
#let reduced = cancel(response)
#let modes = apart(reduced, s)
#let residual = together(sub(modes, reduced))

$
H(s) &= #to-tyrst(response) \
H_"reduced"(s) &= #to-tyrst(reduced) \
               &= #to-tyrst(modes)
$ \
verification: $ #to-tyrst(residual) $
```

$$H(s) = (3 + s)(5 + 2s)(6 + 11s + 6s^2 + s^3)^{-1}$$

$$H_{\text{reduced}(s)} = (5 + 2s)(2 + 3s + s^2)^{-1}$$

$$= 3(1 + s)^{-1} - (2 + s)^{-1}$$

verification:

0

The two terms expose poles at $s = -1$ and $s = -2$ separately, which is often the useful form for inverse transforms or modal reasoning. `together` returns a zero residual after recombination. The cancellation does hide the original restriction $s \neq -3$; keep that restriction when the domain matters.

3.4 Solve a nonlinear system exactly and numerically

The circle $x^2 + y^2 = 25$ and the line $x - y = 1$ meet twice. The exact solver should find both intersections; the numerical solver should find the one nearest its starting point.

```
#let x = symbol("x")
#let y = symbol("y")
#let system = (
  math($x^2 + y^2 - 25$),
  math($x - y - 1$),
)

#let exact = solve(system, (x, y), domain: "real")
#let positive = nsolve-system(
  system, (x, y), (3.0, 3.0), prec: 1e-10,
)
#let negative = nsolve-system(
  system, (x, y), (-3.0, -3.0), prec: 1e-10,
)
#let checks = evaluate-many(
  system, (x, y), (positive, negative),
)
```

exact branches:

1. $x = 4, y = 3$

2. $x = -3, y = -4$

seed (3,3) $\rightarrow$ (4, 3); maximum residual 0.0

seed (-3,-3) $\rightarrow$ (-3, -4); maximum residual 3.552713678800501e-15

The exact calculation finds both branches. Starting near either intersection selects that numerical branch, and the small residual confirms that the point lies on both curves. A poor starting point may still converge elsewhere—or not at all.

3.5 Recover an interpolating polynomial with matrices

Find the quadratic through $(0, 1)$, $(1, 3)$, and $(2, 8)$. Writing $p(t) = a_0 + a_1t + a_2t^2$ turns interpolation into the exact matrix equation $Aa = b$.

```
#let t = symbol("t")
#let A = matrix((
  (1, 0, 0),
  (1, 1, 1),
  (1, 2, 4),
))
#let b = vec((1, 3, 8))
#let coefficients = matrix-solve(A, b)
#let check = matrix-mul(A, coefficients)
#let exact = matrix-is-zero(matrix-sub(check, b))
#let polynomial = add(
  matrix-at(coefficients, 0, 0),
  mul(matrix-at(coefficients, 1, 0), t),
  mul(matrix-at(coefficients, 2, 0), pow(t, 2)),
)

$ A = #to-tyrst(A), quad b = #to-tyrst(b) $ \
$ op("det")(A) = #to-tyrst(det(A)) $ \
$ a = #to-tyrst(coefficients) $ \
$ p(t) = #to-tyrst(polynomial) $ \
$ A a = #to-tyrst(check) $ \
verification: residual matrix is exactly zero: #exact
```

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 3 \\ 8 \end{pmatrix}$$

$$\det(A) = 2$$

$$a = \begin{pmatrix} 1 \\ \frac{1}{2} \\ \frac{3}{2} \end{pmatrix}$$

$$p(t) = 1 + \frac{1}{2}t + \frac{3}{2}t^2$$

$$Aa = \begin{pmatrix} 1 \\ 3 \\ 8 \end{pmatrix}$$

verification: residual matrix is exactly zero: true

The nonzero determinant tells us the coefficients are unique. Reading them back gives the polynomial, and multiplying Aa recovers the three original measurements.

3.6 Map a gradient across a grid

Consider the quadratic surface $f(x, y) = x^2 + xy + y^2$. We first derive its two gradient components, then sample the height and gradient together on a small Cartesian grid.

```
#let x = symbol("x")
#let y = symbol("y")
#let f = math($x^2 + x y + y^2$)
#let fx = derivative(f, x)
#let fy = derivative(f, y)
#let grid = evaluate-grid(
  (f, fx, fy),
  (x, y),
  (
    domain(-1, 1, samples: 3),
    domain(-1, 1, samples: 3),
  ),
),
```

```
)
$ f = #to-typst(f) $
$ partial_x f = #to-typst(fx), quad partial_y f = #to-typst(fy) $
```

$$f = xy + x^2 + y^2$$

$$\partial_x f = 2x + y, \quad \partial_y f = x + 2y$$

x	y	f	$\partial_x f$	$\partial_y f$
-1	-1	3	-3	-3
-1	0	1	-2	-1
-1	1	1	-1	1
0	-1	1	-1	-2
0	0	0	0	0
0	1	1	1	2
1	-1	1	1	-1
1	0	1	2	1
1	1	3	3	3

The table makes the geometry visible: the gradient vanishes at the origin and points uphill everywhere else. `evaluate-grid` pairs each point with one value for every requested expression; use `evaluate-many` when your sample points do not form a Cartesian product.

4 What to expect

Tymbolica deliberately presents a smaller surface than Symbolica itself. The parts covered in this manual work well for exact algebra in documents, but a few boundaries are worth knowing before you choose an approach:

- Symbolic integration is intentionally a separate concern. The companion `tymbolica-rubi` package accepts and returns the same portable Atom payloads. Matrix payloads remain separate from Atom payloads.
- Exact system solving is intended for linear and polynomial equations. Numerical solving depends on a starting point and gives an approximate answer rather than a derivation of convergence.
- The current upstream Symbolica build has a known regression: some exact algebraic solves, including $x - \sqrt{2} = 0$, can hang during compilation. This release does not fix that regression.
- Decimal literals remain floating-point values. A current upstream Wasm bug can mis-evaluate them inside analytic functions. Write exact fractions for those inputs—for example, `cos(1/2)`—and apply `to-float` to the result.
- Matrix entries must be rational-polynomial expressions, and their dimensions must agree for the requested operation.
- Algebraic transformations do not keep a separate list of assumptions. If a manipulation is valid only under a condition such as $x \neq 1$, preserve that condition in the surrounding document.
- Some unusual Typst math structures may not parse. `array-tree` can help show what the parser received.

For symbolic integration, use `tymbolica-rubi`. For arbitrary precision or deeper polynomial algorithms, use Symbolica directly.

4.1 When something looks wrong

Symptom	Likely cause	What to try
expected content, found bytes	A symbolic result was inserted directly into <code>\$...\$</code> .	Display it with <code>to-typst</code> .

Symptom	Likely cause	What to try
A symbolic result fails in another API	The bytes hold a matrix or came from an incompatible package version.	Use the matching package version and keep matrix values in matrix APIs.
A derivative or series leaves a function unchanged	<code>sin</code> , <code>cos</code> , or another analytic function was read as an ordinary name.	Use <code>init(namespace: "symbolica")</code> for Symbolica built-ins.
An analytic function of a decimal gives an unexpected value	Its floating-point argument encountered the current upstream Wasm bug.	Use an exact fraction such as <code>cos(1/2)</code> , then call <code>to-float</code> .
A replacement does not match	Repeated wildcards would have to capture different expressions.	Check that every occurrence of the wildcard should match the same value.
An exact solver errors	An equation was not rearranged to zero, or the system is outside the supported polynomial scope.	Move every term to the left; try <code>nsolve-system</code> for a numerical branch.
Compilation hangs in an exact algebraic solve	The current upstream regression can affect even $x - \sqrt{2} = 0$.	Cancel compilation. This case needs an upstream fix; see the repository's <code>repros/algebraic-sqrt2</code> .
A numerical solver errors or finds an unwanted root	The starting point leads to a different root or no root.	Try another physically meaningful initial guess and check residuals.
A matrix operation errors	Shapes differ, a matrix is singular, or an entry is unsupported.	Inspect <code>matrix-shape</code> , <code>det</code> , and the coefficient expressions.

5 API reference

The worked chapters are meant for reading; this section is meant for looking things up. The generated groups below cover the complete top-level API.

5.1 Parsing, symbols, and rendering

5.1.1 math

Parse Typst math content into an opaque Symbolica atom payload.

Arithmetic, fractions, powers, roots, absolute values, calls, and common Typst math structures are translated through the configured grammar. Matrix-valued `mat(...)` and `vec(...)` content must instead be passed to `matrix` or `vec`. Keep the returned bytes opaque and use this module's functions to inspect or transform them. Decimal literals remain floating-point coefficients; write a fraction when you need exact input.

```
#let expr = math($x + 1$)
#to-typst(expr)
```

 $1 + x$

Parameters

```
math(
  eqn: content,
  grammar: dictionary none,
  namespace: str none
) -> bytes
```

eqn `content`

Math content, normally written as `$...$`.

grammar `dictionary` or `none`

Parser grammar override. none uses the default engine grammar.

Default: `none`

namespace `str` or `none`

Namespace for parsed symbols. none uses the engine namespace ("typst" for the top-level function).

Default: `none`

5.1.2 atom

Convert a supported Typst value or math expression into an atom payload.

Existing atom bytes pass through unchanged. Content is parsed like math; integers become exact numbers and floats remain floating-point coefficients. Strings are parsed as leaf values: numeric strings become numbers and other strings become symbols in the engine's namespace. Matrix payloads are not atom payloads and should not be passed to atom-only algebra functions.

```
#to-typst(atom("x"))
```

 x

Parameters

```
atom(value: bytes content int float str) -> bytes
```

value bytes or `content` or `int` or `float` or `str`

Value to convert.

5.1.3 symbol

Construct a named Symbolica symbol with semantic Typst metadata.

Unlike `wild`, this is an ordinary mathematical symbol and therefore does not capture subexpressions during pattern matching. The returned content can be interpolated into native Typst math; `math` reads its name and namespace from versioned metadata instead of guessing from its appearance. Tags travel with that metadata for higher-level notation layers but do not change Symbolica's algebraic behavior.

```
#let x = symbol("x", namespace: "model", tags:
  ("model::positive",))
#to-typst(math($#x^2 + 1$))
```

 $1 + x^2$

Parameters

```
symbol(
  name: str,
  namespace: str none,
  tags: array
) -> content
```

name `str`

Symbol name without a namespace prefix or wildcard suffix.

namespace `str` or `none`

Namespace override. `none` uses the engine namespace.

Default: `none`

tags `array`

Portable semantic labels retained in the attached metadata.

Default: `()`

5.1.4 function

Construct a callable Symbolica function with semantic Typst metadata.

Calling the returned Typst function constructs the exact Symbolica call, prints that Atom, and attaches one metadata envelope to the complete call. Interpolate callable bindings inside `math`, for example `#f(x)`.

```
#let f = function("f", namespace: "model",
tags: ("model::smooth",))
#to-typst(math($#f(symbol("x")) + 1$))
```

$$1 + f(x)$$

Parameters

```
function(
  name: str,
  namespace: str none,
  tags: array
) -> function
```

name `str`

Function-head name.

namespace `str` or `none`

Namespace override. `none` uses the engine namespace.

Default: `none`

tags `array`

Portable semantic labels retained in the attached metadata.

Default: `()`

5.1.5 wild

Construct a Symbolica pattern wildcard.

This creates the symbol name followed by `level` underscores. A wildcard is a pattern placeholder used by `rule`, `replace`, and `replace-wildcards`; it is not an ordinary unknown for algebra or solving. Use `symbol` for an ordinary mathematical symbol.

```
#raw(canonical(wild("a")))
```

```
a_
```

Parameters

```
wild(
  name: str,
  level: int,
  namespace: str or none
) -> bytes
```

name `str`

Base name of the wildcard, without trailing underscores.

level `int`

Number of underscore levels to append. 1 creates a conventional single wildcard such as `a_`; 0 appends no underscore and therefore creates an ordinary symbol instead.

Default: `1`

namespace `str` or `none`

Namespace override. `none` uses the engine namespace.

Default: `none`

5.1.6 array-tree

Render the parse tree for a Typst math expression.

This is a diagnostic view of the tree consumed by `math`; it does not create a Symbolica atom.

```
#array-tree($(x + 1)^2$)
```

```
(
  strong(body: raw(text: "pow")),
  (
    styled(child: raw(text: "base"), ..),
    (
      strong(body: raw(text: "()")),
      (
        styled(child: raw(text: "expr"), ..),
        (
          strong(body: raw(text: "add")),
          equation(body: [x]),
          equation(body: [1]),
        ),
      ),
    ),
  ),
  (
    styled(child: raw(text: "exp"), ..),
    equation(body: [2]),
  ),
)
```

Parameters

```
array-tree(
  eqn: content,
  grammar: dictionary none
) -> content
```

eqn `content`

Math content to inspect.

grammar `dictionary` or `none`

Parser grammar override. none uses the engine grammar.

Default: `none`

5.1.7 canonical

Render an atom or matrix payload as Symbolica source text.

```
#raw(canonical(math($x + 1$)))
```

```
1+x
```

Parameters

```
canonical(
  expr: bytes content int float str,
  namespaces: bool
) -> str
```

expr `bytes` or `content` or `int` or `float` or `str`

Atom or matrix payload. Other supported expression values are first converted as by atom.

namespaces `bool`

Include symbol namespaces in the output.

Default: `false`

5.1.8 notation

Build a document-side Atom notation layer.

Exact heads entries change one namespaced symbol. Exact calls entries receive the complete function node and its arguments. tags and classes provide broader fallbacks. Renderer closures receive a context dictionary; exact Atom metadata is attached by to-typst after a renderer returns.

Parameters

```
notation(
  heads,
  calls,
  tags,
  classes,
  fallback-head,
```

```

    fallback-call,
    fallback-node,
    fallback
) -> dictionary

```

5.1.9 merge-notation

Merge notation layers from least to most specific.

Parameters

```
merge-notation(..layers) -> dictionary
```

5.1.10 to-typst-source

Render an atom or matrix payload as Typst math source.

```
#raw(to-typst-source(math($x + 1$)))
```

 $1+x$

Parameters

```
to-typst-source(expr: bytes | content | int | float | str) -> str
```

expr bytes or content or int or float or str

Atom or matrix payload, or a supported expression value.

5.1.11 to-typst

Render an atom or matrix payload as evaluated Typst math content.

Symbolica exports a generic algebra tree; Typst constructs its visual math directly from that tree. Ordinary calls retain an exact annotated head and visible arguments. A custom full-call renderer receives the complete node, and the framework attaches the exact call metadata after it returns. Matrix output remains display-only.

```
#to-typst(math($x + 1$))
```

 $1+ x$

Parameters

```

to-typst(
  expr: bytes | content | int | float | str,
  notation: dictionary | none,
  block: bool
) -> content

```

expr bytes or content or int or float or str

Atom or matrix payload, or a supported expression value.

notation dictionary or none

Per-call document notation layered over the engine's default. none uses that default unchanged.

Default: `none`

block `bool`

Render as a block equation instead of inline math.

Default: `false`

5.1.12 to-latex

Render an atom or matrix payload as LaTeX source.

```
#raw(to-latex(math($x^2 + 1$)))
```

$$1+x^2$$

Parameters

`to-latex`(`expr`: `bytes` `content` `int` `float` `str`) -> `str`

expr `bytes` or `content` or `int` or `float` or `str`

Atom or matrix payload, or a supported expression value.

5.1.13 to-float

Approximate numerical coefficients and built-in functions as decimals.

Variables remain symbolic. The WebAssembly build supports between 1 and 16 significant decimal digits. Use `evaluate` on the original expression when you need a numerical value rather than a symbolic display form. Expressions containing built-in calls with very large numeric arguments remain exact.

```
#to-tyrst(to-float(math($1/3$), decimal-prec: 6))
```

$$3.3333$$

Exact rational arguments to Symbolica built-ins can be approximated too:

```
#let sym = init(namespace: "symbolica")
#let parse = sym.math
#let approximate = sym.to-float
#let render = sym.to-tyrst
#render(approximate(parse($cos(1/2$)), decimal-prec: 6))
```

$$8.7758$$

Parameters

`to-float`(
`expr`: `bytes` `content` `int` `float` `str`,
`decimal-prec`: `int`
) -> `bytes`

expr `bytes` or `content` or `int` or `float` or `str`

Expression to approximate.

decimal-prec `int`

Number of significant decimal digits to retain, from 1 through 16.

Default: `16`

5.2 Algebra and calculus

5.2.1 simplify

Re-import and re-export an atom in Symbolica's canonical internal form.

This operation does not perform algebraic simplification; it currently acts as a normalization and validation round-trip for atom payloads.

```
#to-typst(simplify(math($x + 0$)))
```

 x

Parameters

```
simplify(expr: bytes content int float str) -> bytes
```

expr bytes or content or int or float or str

Atom payload or supported expression value to normalize.

5.2.2 expand

Expand an expression through Symbolica's polynomial expansion.

```
#to-typst(expand(math($(x + 1)^2$)))
```

 $1 + 2$
 $x + x^2$

Parameters

```
expand(expr: bytes content int float str) -> bytes
```

expr bytes or content or int or float or str

Atom payload or supported expression value to expand.

5.2.3 factor

Factor an expression exactly.

```
#to-typst(factor(math($x^2 + 2 x + 1$)))
```

 $(1 + x)^2$

Parameters

```
factor(
  expr: bytes content int float str,
  complex: bool,
  square-free: bool
) -> bytes
```

expr bytes or content or int or float or str

Atom payload or supported expression value to factor.

complex bool

Factor over the complex rationals instead of the rationals. Cannot be combined with square-free.

Default: false

square-free bool

Return a square-free factorization, preserving multiplicities without fully splitting every factor. Complex input is handled over the complex rationals. Cannot be combined with complex.

Default: false

5.2.4 together

Write a rational expression over a common denominator.

```
#to-tyrst(together(math($1/x + 1/y$)))
```

$$x^{-1}$$

$$y^{-1}$$

$$(x + y)$$

Parameters

together(expr: bytes content int float str) -> bytes

expr bytes or content or int or float or str

Expression whose rational terms should be combined.

5.2.5 cancel

Cancel common factors between numerators and denominators.

Parts of the expression without a cancellation are left alone. Remember that canceling a factor can hide a point excluded by the original formula.

```
#to-tyrst(cancel(math($x^2 - 1)/(x - 1$)))
```

$$1 + x$$

Parameters

cancel(expr: bytes content int float str) -> bytes

expr bytes or content or int or float or str

Rational expression in which to cancel common factors.

5.2.6 apart

Decompose a rational expression into partial fractions.

Denominators are decomposed in the given indeterminate.

```
#let x = symbol("x")
#to-tyrst(apart(math(2 x + 3)/((x + 1)(x + 2)), x))
```

$$(1 + x)^{-1} + (2 + x)^{-1}$$

Parameters

```
apart(
  expr: bytes | content | int | float | str,
  var: bytes | content | str
) -> bytes
```

expr bytes or content or int or float or str

Rational expression to decompose.

var bytes or content or str

Indeterminate for the decomposition.

5.2.7 collect

Collect terms by powers of one or more variables or functions.

```
#let x = symbol("x")
#to-tyrst(collect(math(5 x + x y + x^2 + 5), x))
```

$$5 + x$$

$$(5 + y) + x^2$$

Parameters

```
collect(
  expr: bytes | content | int | float | str,
  variables: bytes | content | str | array
) -> bytes
```

expr bytes or content or int or float or str

Expression whose terms should be collected.

variables bytes or content or str or array

One variable or function, or an array of them.

5.2.8 coefficient

Extract the coefficient of a literal monomial or subexpression.

For example, asking for the coefficient of x^2 in $5x + xy + x^2 + yx^2$ returns $1 + y$.

```
#let x = symbol("x")
#to-tyrst(coeffcient(math($5 x + x y + x^2 + y
x^2$), pow(x, 2)))
```

 $1 + y$

Parameters

```
coeffcient(
  expr: bytes content int float str,
  monomial: bytes content int float str
) -> bytes
```

expr bytes or content or int or float or str

Expression to inspect.

monomial bytes or content or int or float or str

Literal monomial or subexpression whose coefficient is wanted.

5.2.9 coefficient-list

Return collected key-coefficient pairs for one or more indeterminates.

Each result is (key, coefficient), both as atom payloads. A key of 1 carries terms not polynomially collected in the requested variables. A coefficient that vanishes only through a deeper identity may remain.

```
#let x = symbol("x")
#let pairs = coeffcient-list(math($x^2 + 5 x +
7$), x)
#pairs.map(pair => [#to-tyrst(pair.at(0)): #to-
tyrst(pair.at(1))]).join[, ]
```

 $1: 7, x: 5, x^2: 1$

Parameters

```
coeffcient-list(
  expr: bytes content int float str,
  variables: bytes content str array
) -> array
```

expr bytes or content or int or float or str

Expression to inspect.

variables bytes or content or str or array

One variable or function, or an array of them.

5.2.10 terms

Return the top-level additive terms of an expression.

This does not expand or recurse. A value that is not a sum is returned as a one-element array.

```
#terms(math($x^2 + 2 x + 1$)).map(to-
typst).join[, ]
```

1, 2

x , x^2

Parameters

```
terms(expr: bytes content int float str) -> array
```

expr bytes or content or int or float or str

Expression to split into top-level summands.

5.2.11 indeterminates

Return the variables and function expressions that act as indeterminates.

Results are sorted in Symbolica's internal order.

```
#indeterminates(math($f(x) + y$)).map(to-
typst).join[, ]
```

x , y , $f(x)$

Parameters

```
indeterminates(
  expr: bytes content int float str,
  enter-functions: bool
) -> array
```

expr bytes or content or int or float or str

Expression to inspect.

enter-functions bool

Traverse function arguments as well as collecting the function itself.

Default: **true**

5.2.12 contains

Test whether an expression literally contains another expression.

The test follows the stored expression structure: for example, $x*y*z$ contains x , but does not contain the regrouped product $x*y$ as a node.

```
#contains(math($x y z$), symbol("x"))
```

true

Parameters

```
contains(
  expr: bytes content int float str,
  subexpression: bytes content int float str
) -> bool
```

expr bytes or content or int or float or str

Expression to search.

subexpression bytes or content or int or float or str

Literal subexpression to look for.

5.2.13 is-constant

Test whether an expression has no user-defined variables or functions.

Supported built-in functions such as sin and cos remain constant when all of their arguments are constant.

```
#is-constant(math(cos(2) + 1/3))
```

```
true
```

Parameters

`is-constant(expr: bytes content int float str) -> bool`

expr bytes or content or int or float or str

Expression to test.

5.2.14 derivative

Differentiate an expression exactly with respect to an indeterminate.

```
#let x = symbol("x")
#to-tyrst(derivative(math((x + 1)^2), x))
```

```
2
```

```
(1 + x)
```

Parameters

```
derivative(
  expr: bytes content int float str,
  var: bytes content str
) -> bytes
```

expr bytes or content or int or float or str

Expression to differentiate.

var bytes or content or str

Symbolica indeterminate, normally created with `symbol`.

5.2.15 series

Compute a univariate series expansion around expansion-point.

The truncation depth is the rational number depth / depth-denom. With an absolute depth it is measured directly in var; with a relative depth it is measured from the lowest order encountered in the expression.

```
#let sym = init(namespace: "symbolica")
#let m = sym.math
#let v = sym.symbol
#let ser = sym.series
#let render = sym.to-tyrst
#render(ser(m($cos(x)/(x + 1)$), v("x"), 0, 3))
```

$$1 - x + \frac{1}{2}x^2 - \frac{1}{2}x^3$$

Parameters

```
series(
  expr: bytes content int float str,
  var: bytes content str,
  expansion-point: bytes content int float str,
  depth: int,
  depth-denom: int,
  depth-is-absolute: bool
) -> bytes
```

expr bytes or content or int or float or str

Expression to expand.

var bytes or content or str

Expansion variable, normally created with symbol.

expansion-point bytes or content or int or float or str

Point about which to expand.

depth int

Numerator of the requested truncation depth.

depth-denom int

Denominator of the truncation depth.

Default: 1

depth-is-absolute bool

Use an absolute depth when true, or a depth relative to the lowest order in the expression when false.

Default: true

5.3 Rewriting

5.3.1 rule

Build a reusable replacement rule for replace-multiple.

pattern and rhs may contain symbols created with wild. The returned dictionary contains opaque atom payloads plus the matching options; pass it to replace-multiple rather than editing it manually.

```
#let r = rule(math($f("a_")$), math($g("a_")$))
#to-tyrst(replace-multiple(math($f(x)$), (r,)))
```

 $g(x)$

Parameters

```
rule(
  pattern: bytes content int float str,
  rhs: bytes content int float str,
  non-greedy-wildcards: array,
  min-level: int,
  max-level: int none,
  level-range: array none,
  level-is-tree-depth: bool,
  partial: bool,
  allow-new-wildcards-on-rhs: bool,
  rhs-cache-size: int
) -> dictionary
```

pattern bytes or content or int or float or str

Pattern to match.

rhs bytes or content or int or float or str

Expression substituted for each match. Wildcards captured by pattern are substituted in this expression.

non-greedy-wildcards array

Wildcards that should prefer the smallest possible match.

Default: ()

min-level int

Lowest expression level at which matching is allowed; the root is level zero.

Default: 0

max-level int or none

Highest allowed matching level, or none for no upper bound.

Default: none

level-range array or none

Optional (minimum, maximum) pair overriding min-level and max-level; the maximum may be none.

Default: none

level-is-tree-depth bool

Count full expression-tree depth when true; otherwise levels increase when entering functions.

Default: `false`

partial `bool`

Allow a pattern to match part of a sum, product, or other term instead of requiring the whole term.

Default: `true`

allow-new-wildcards-on-rhs `bool`

Permit wildcards on rhs that do not occur in pattern. When false, such rules are rejected.

Default: `false`

rhs-cache-size `int`

Maximum number of substituted right-hand sides cached; use zero to disable this cache.

Default: `100`

5.3.2 replace

Replace subexpressions matching pattern with rhs.

By default all non-overlapping outermost matches are replaced once. `repeat: true` reapplies the rule until the expression stops changing; rules that cycle therefore do not terminate. Use rule plus `replace-multiple` when several patterns must be considered together.

```
#to-tyrst(replace(math($f(x, y)$),
math($f("a_", "b_")$), math($g("b_", "a_")$)))
```

$$g(y, x)$$

Parameters

```
replace(
  expr: bytes content int float str,
  pattern: bytes content int float str,
  rhs: bytes content int float str,
  repeat: bool,
  once: bool,
  bottom-up: bool,
  nested: bool,
  non-greedy-wildcards: array,
  min-level: int,
  max-level: int none,
  level-range: array none,
  level-is-tree-depth: bool,
  partial: bool,
  allow-new-wildcards-on-rhs: bool,
  rhs-cache-size: int
) -> bytes
```

expr bytes or content or int or float or str

Expression in which to replace matches.

pattern bytes or content or int or float or str

Pattern to match.

rhs bytes or content or int or float or str

Replacement expression.

repeat bool

Reapply the rule until it makes no further change.

Default: false

once bool

Replace only the first match found during a pass.

Default: false

bottom-up bool

Visit deepest matches before outer matches.

Default: false

nested bool

Replace nested matches from the deepest outward, acting on the result of each inner replacement.

Default: false

non-greedy-wildcards array

Wildcards that should prefer the smallest possible match.

Default: ()

min-level int

Lowest allowed matching level; the root is level zero.

Default: 0

max-level int or none

Highest allowed matching level, or none for no upper bound.

Default: none

level-range array or none

Optional (minimum, maximum) pair overriding min-level and max-level.

Default: none

level-is-tree-depth `bool`

Count full tree depth instead of function-entry depth.

Default: `false`

partial `bool`

Allow matching a part of a term rather than the entire term.

Default: `true`

allow-new-wildcards-on-rhs `bool`

Permit rhs wildcards that are absent from pattern.

Default: `false`

rhs-cache-size `int`

Maximum number of substituted right-hand sides cached; zero disables the cache.

Default: `100`

5.3.3 replace-multiple

Apply several reusable replacement rules together.

The traversal options apply to the combined rule set. With `repeat: true`, the complete set is reapplied until no rule changes the expression; cyclic rule sets do not terminate.

```
#let r1 = rule(math($f("a_")$),
math($h("a_")$))
#let r2 = rule(symbol("x"), symbol("z"))
#to-tyrst(replace-multiple(math($f(x) + x$),
(r1, r2)))
```

$$z + h(x)$$

Parameters

```
replace-multiple(
  expr: bytes content int float str,
  rules: array,
  repeat: bool,
  once: bool,
  bottom-up: bool,
  nested: bool
) -> bytes
```

expr bytes or content or int or float or str

Expression in which to replace matches.

rules array

Array of dictionaries returned by rule.

repeat bool

Reapply the rule set until it makes no further change.

Default: `false`

once bool

Replace only the first match found during a pass.

Default: `false`

bottom-up bool

Visit deepest matches before outer matches.

Default: `false`

nested bool

Replace nested matches from deepest to outermost, acting on intermediate results.

Default: `false`

5.3.4 replace-wildcards

Substitute explicit values for wildcard placeholders in a pattern.

This does not search another expression. It transforms pattern itself and requires each key in replacements to be a wildcard symbol.

```
#to-tyrst(replace-wildcards(math($k("a_")$),
((wild("a"), math($x + 1$))),))
```

$$k(1 + x)$$

Parameters

```
replace-wildcards(
  pattern: bytes content int float str,
  replacements: array
) -> bytes
```

pattern bytes or content or int or float or str

Pattern containing wildcards to substitute.

replacements array

Array of (wildcard, replacement) pairs.

5.4 Evaluation and solving

5.4.1 evaluate

Evaluate one expression numerically with optional substitutions.

values maps atom keys to real numbers or complex dictionaries of the form (re: number, im: number). Evaluation must eliminate every unsupported symbolic quantity. The result always has the exact shape (re: float, im: float), even when it is real.

```
#let x = symbol("x")
#let y = symbol("y")
#repr(evaluate(math($x^2 + y$), values: ((x,
2.0), (y, 3.0))))
```

```
(re: 7.0, im: 0.0)
```

Parameters

```
evaluate(
  expr: bytes content int float str,
  values: array
) -> dictionary
```

expr bytes or content or int or float or str

Expression to evaluate.

values array

Array of (expression, value) substitution pairs. Values may be real numbers or (re: ..., im: ...) dictionaries.

Default: ()

5.4.2 domain

Describe one real sampling axis for evaluate-grid.

Both endpoints are included when samples is greater than one. A single-sample domain evaluates only at min. The returned dictionary has the exact shape (min: float, max: float, samples: int).

```
#repr(domain(-1, 1, samples: 3))
```

```
(min: -1.0, max: 1.0, samples: 3)
```

Parameters

```
domain(
  min: int float,
  max: int float,
  samples: int
) -> dictionary
```

min int or float

Finite lower endpoint; it must be less than max.

max int or float

Finite upper endpoint.

samples int

Number of evenly spaced points. Must be positive.

Default: 200

5.4.3 evaluate-many

Evaluate one or more expressions at explicit points in one batch.

A single expression or variable may be passed directly; otherwise use an array. Every point must supply one real or complex value per variable. The result has one row per point in input order and one (re: float, im: float) dictionary per expression in expression order.

```
#let x = symbol("x")
#let y = symbol("y")
#let rows = evaluate-many((math($x + y$),
math($x y$)), (x, y), ((1, 2), (3, 4)))
#repr(rows)
```

```
(
  ((re: 3.0, im: 0.0), (re: 2.0, im: 0.0)),
  ((re: 7.0, im: 0.0), (re: 12.0, im: 0.0)),
)
```

Parameters

```
evaluate-many(
  expressions: bytes content array int float str,
  variables: bytes content array str,
  points: array
) -> array
```

expressions bytes or content or array or int or float or str

One expression or a non-empty array of expressions to evaluate.

variables bytes or content or array or str

One variable or an array defining input-column order.

points array

Array of input rows. Each row is an array ordered like variables; for one variable, a scalar point is also accepted.

5.4.4 evaluate-grid

Evaluate expressions over a Cartesian product of real domains in one batch.

The result is (shape: array, points: array, values: array). shape lists the sample count of every domain. The grid axes are flattened into rows with the last domain varying fastest: each points row contains real coordinates in variable order, while the corresponding values row contains one (re: float, im: float) dictionary per expression.

```
#let x = symbol("x")
#let y = symbol("y")
#let grid = evaluate-grid(math($x^2 + y$), (x,
y), (domain(-1, 1, samples: 3), domain(0, 1,
samples: 2)))
shape: #repr(grid.shape); values:
#repr(grid.values)
```

```
shape: (3, 2) values: (
  ((re: 1.0, im: 0.0),),
  ((re: 2.0, im: 0.0),),
  ((re: 0.0, im: 0.0),),
  ((re: 1.0, im: 0.0),),
  ((re: 1.0, im: 0.0),),
  ((re: 2.0, im: 0.0),),
)
```

Parameters

```
evaluate-grid(
  expressions: bytes content array int float str,
  variables: bytes content array str,
  domains: array
) -> dictionary
```

expressions bytes or content or array or int or float or str

One expression or a non-empty array of expressions.

variables bytes or content or array or str

One variable or an array defining grid-axis order.

domains array

One domain dictionary per variable, in the same order.

5.4.5 solve

Solve a linear or supported polynomial nonlinear system exactly.

Each expression in system is understood to equal zero. Polynomial systems are solved exactly through Symbolica's Gröbner-basis and algebraic-root machinery; coefficients may contain symbolic parameters when Symbolica can treat them rationally. domain may be "complex", "real", "rational", or "integer". The domain also supplies local assumptions for unrestricted external parameters: for example, "real" treats them as real while solving, without changing their global symbol properties.

The result is a dictionary with branches, the requested variables, inferred external parameters, and domain. coverage is "complete" or "generic": generic results cover only parameter values where every expression in coverage-guard is nonzero. Other parameter values may have additional solutions. An empty branches array proves there are no solutions only when coverage is complete.

Each branch has a values array in the requested variable order. Free variables appear as themselves in that array and are also listed in free-variables. conditions retain restrictions: "zero", "nonzero", and "positive" carry an expression; "domain-membership" carries a variable, value, and domain. conditional reports whether these restrictions remain; point means there are neither free variables nor branch conditions. The result's coverage guard still applies even when a branch is a point.

Branches also provide domain, dimension, and codimension (the number of requested variables minus the dimension). The result's dimension is the largest branch dimension, or -1 for the complete empty set. Dimensions are none when Symbolica cannot establish them; they are not inferred by counting free variables.

```
#let x = symbol("x")
#let y = symbol("y")
#let solutions = solve((math($x^2 - 1$),
math($y - x$)), (x, y), domain: "real")
#repr(solutions.branches.map(solution =>
solution.values.map(canonical)))
```

```
((("1", "1"), ("-1", "-1")))
```

Parameters

```
solve(
  system: array content int float str,
  variables: array content str,
  domain: str
) -> dictionary
```

system array or content or int or float or str

Expressions understood to equal zero.

variables array or content or str

Variables to solve for, in values order.

domain str

Exact solution domain: "complex", "real", "rational", or "integer".

Default: "complex"

5.4.6 nsolve

Find a real root of a univariate expression with Newton's method.

The expression is interpreted as equal to zero and evaluated with f64 arithmetic. Convergence is local and depends on init; failure to converge within max-iterations produces an error.

```
#let x = symbol("x")
#repr(nsolve(math($x^2 - 2$), x, 1.0))
```

```
1.4142135623746899
```

Parameters

```
nsolve(
  expr: bytes content int float str,
  var: bytes content str,
  init: int float,
  prec: int float,
  max-iterations: int
) -> float
```

expr bytes or content or int or float or str

Expression understood to equal zero.

var bytes or content or str

Real solve variable, normally created with symbol.

init `int` or `float`

Initial real guess.

prec `int` or `float`

Numerical tolerance for the Newton iteration.

Default: `1e-4`

max-iterations `int`

Maximum number of Newton iterations.

Default: `1000`

5.4.7 nsolve-system

Find a common real root of a system with multivariate Newton iteration.

Every expression is interpreted as equal to zero. `variables` and `init` must have matching lengths, and the returned floats follow that same order. Convergence is local and is not guaranteed for an arbitrary initial guess.

```
#let x = symbol("x")
#let y = symbol("y")
#repr(nsolve-system((math($x^2 + y - 3$),
math($x - y$)), (x, y), (1.0, 1.0)))
```

```
(1.3027756377319948, 1.3027756377319948)
```

Parameters

```
nsolve-system(
  system: array content int float str,
  variables: array content str,
  init: array,
  prec: int float,
  max-iterations: int
) -> array
```

system `array` or `content` or `int` or `float` or `str`

Expressions understood to equal zero.

variables `array` or `content` or `str`

Variables to solve for, in result order.

init `array`

Initial real value for every variable.

prec `int` or `float`

Numerical tolerance for the Newton iteration.

Default: **1e-4**

max-iterations **int**

Maximum number of Newton iterations.

Default: **1000**

5.5 Matrices

5.5.1 matrix

Convert Typst values into an opaque Symbolica matrix payload.

Accepted forms are Typst math `mat(...)` or `vec(...)`, a non-empty rectangular nested array, a flat array interpreted as a column vector, a scalar interpreted as a one-by-one matrix, or an existing payload. Every entry must be convertible to a rational polynomial; general transcendental expressions are not valid matrix entries.

```
#to-tpst(matrix($mat(1, 2; 3, 4)$))
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Parameters

`matrix(value: bytes content array int float str) -> bytes`

value bytes or content or array or int or float or str

Matrix source value.

5.5.2 vec

Convert a non-empty sequence into a column-vector matrix payload.

Array entries must be rational-polynomial compatible. Typst math `vec(...)` is accepted directly; existing matrix bytes pass through unchanged.

```
#to-tpst(vec((1, 2)))
```

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Parameters

`vec(values: array content bytes) -> bytes`

values array or content or bytes

Vector entries, Typst `vec(...)` content, or an existing matrix payload.

5.5.3 identity

Create an n by n identity matrix payload.

```
#to-tpst(identity(2))
```

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Parameters

`identity(n: int) -> bytes`

n `int`

Positive matrix dimension.

5.5.4 `eye`

Create a square diagonal matrix from a non-empty entry array.

Every entry must be rational-polynomial compatible.

```
#to-tyrst(eye((1, 2)))
```

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

Parameters

`eye(diag: array) -> bytes`

diag `array`

Diagonal entries in top-left to bottom-right order.

5.5.5 `matrix-add`

Add two matrices entry by entry.

The matrices must have equal shapes. Matrix source values accepted by `matrix` may be supplied directly.

```
#to-tyrst(matrix-add(matrix(((1, 2), (3, 4))),
identity(2)))
```

$$\begin{pmatrix} 2 & 2 \\ 3 & 5 \end{pmatrix}$$

Parameters

```
matrix-add(
  lhs: bytes content array int float str,
  rhs: bytes content array int float str
) -> bytes
```

lhs `bytes` or `content` or `array` or `int` or `float` or `str`

Left matrix.

rhs `bytes` or `content` or `array` or `int` or `float` or `str`

Right matrix of the same shape.

5.5.6 `matrix-sub`

Subtract two matrices entry by entry.

The matrices must have equal shapes. Matrix source values accepted by `matrix` may be supplied directly.

```
#to-tyrst(matrix-sub(matrix(((1, 2), (3, 4))),
identity(2)))
```

$$\begin{pmatrix} 0 & 2 \\ 3 & 3 \end{pmatrix}$$

Parameters

```
matrix-sub(
  lhs: bytes content array int float str,
  rhs: bytes content array int float str
) -> bytes
```

lhs bytes or content or array or int or float or str

Left matrix.

rhs bytes or content or array or int or float or str

Right matrix of the same shape.

5.5.7 matrix-mul

Multiply two matrices, or multiply a matrix by a scalar expression.

Matrix multiplication requires compatible inner dimensions. A scalar must be rational-polynomial compatible. Because opaque bytes are interpreted as matrix payloads in the right-hand position, pass scalar atoms as ordinary values or math content rather than as preconstructed atom bytes.

```
#to-typst(matrix-mul(matrix(((1, 2), (3, 4))),
identity(2)))
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Parameters

```
matrix-mul(
  lhs: bytes content array int float str,
  rhs: bytes content array int float str
) -> bytes
```

lhs bytes or content or array or int or float or str

Left matrix.

rhs bytes or content or array or int or float or str

Right matrix, or scalar Typst value/math content.

5.5.8 matrix-div-scalar

Divide every matrix entry by a nonzero scalar expression.

The scalar must be rational-polynomial compatible.

```
#to-typst(matrix-div-scalar(matrix(((2, 4), (6,
8))), 2))
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Parameters

```
matrix-div-scalar(
  lhs: bytes content array int float str,
```

```
rhs: bytes content int float str
) -> bytes
```

lhs bytes or content or array or int or float or str

Matrix dividend.

rhs bytes or content or int or float or str

Nonzero scalar divisor.

5.5.9 transpose

Transpose a matrix, exchanging rows and columns.

```
#to-tyrst(transpose(matrix(((1, 2), (3, 4)))))
```

$$\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Parameters

```
transpose(matrix: bytes content array int float str) -> bytes
```

matrix bytes or content or array or int or float or str

Matrix to transpose.

5.5.10 det

Compute the exact determinant of a square matrix.

The result is an atom payload rather than a matrix payload.

```
#to-tyrst(det(matrix(((1, 2), (3, 4)))))
```

$$-2$$

Parameters

```
det(matrix: bytes content array int float str) -> bytes
```

matrix bytes or content or array or int or float or str

Square matrix.

5.5.11 inv

Compute the exact inverse of an invertible square matrix.

Singular and non-square matrices produce an error.

```
#to-tyrst(inv(matrix(((1, 2), (3, 4)))))
```

$$\begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Parameters

```
inv(matrix: bytes content array int float str) -> bytes
```

matrix bytes or content or array or int or float or str

Invertible square matrix.

5.5.12 matrix-solve

Solve the matrix equation $A \times b = c$ exactly.

The row counts of A and b must agree. This strict solver reports an error when the system does not have the required unique solution; use `matrix-solve-any` for underdetermined systems.

```
#let A = matrix($mat(2, 1; 1, -1)$)
#let b = vec((5, 1))
#to-typst(matrix-solve(A, b))
```

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Parameters

```
matrix-solve(
  A: bytes content array int float str,
  b: bytes content array int float str
) -> bytes
```

A bytes or content or array or int or float or str

Coefficient matrix.

b bytes or content or array or int or float or str

Right-hand-side matrix or column vector.

5.5.13 matrix-solve-any

Solve $A \times b = c$ exactly, choosing one solution if underdetermined.

The row counts of A and b must agree. An inconsistent system still produces an error.

```
#let A = matrix($mat(2, 1; 1, -1)$)
#let b = vec((5, 1))
#to-typst(matrix-solve-any(A, b))
```

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Parameters

```
matrix-solve-any(
  A: bytes content array int float str,
  b: bytes content array int float str
) -> bytes
```

A bytes or content or array or int or float or str

Coefficient matrix.

b bytes or content or array or int or float or str

Right-hand-side matrix or column vector.

5.5.14 row-reduce

Row-reduce a matrix exactly using Gaussian elimination.

Returns (matrix: bytes, rank: int). By default every column may contain a pivot. Set max-col to limit pivot search to the first max-col columns, which is useful for an augmented matrix whose trailing columns are right-hand sides; all columns are still transformed by the row operations.

```
#let rr = row-reduce(matrix(((1, 2), (3, 4))))
rank #rr.rank: #to-tyrst(rr.matrix)
```

rank 2: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Parameters

```
row-reduce(
  matrix: bytes content array int float str,
  max-col: int none
) -> dictionary
```

matrix bytes or content or array or int or float or str

Matrix to reduce.

max-col int or none

Number of leading columns eligible for pivots. none uses every column. An explicit value must be between zero and the column count.

Default: none

5.5.15 augment

Horizontally concatenate two matrices as [lhs rhs].

Both matrices must have the same number of rows.

```
#to-tyrst(augment(identity(2), matrix(((1, 2),
(3, 4)))))
```

$\begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 4 \end{pmatrix}$

Parameters

```
augment(
  lhs: bytes content array int float str,
  rhs: bytes content array int float str
) -> bytes
```

lhs bytes or content or array or int or float or str

Left block.

rhs bytes or content or array or int or float or str

Right block with the same row count.

5.5.16 split-col

Split a matrix into left and right column blocks.

Returns exactly (left, right), both as matrix payloads. The current matrix backend requires $\text{index} > 0$ and $\text{index} < \text{columns} - 1$.

```
#let aug = augment(identity(2), matrix(((1, 2),
(3, 4))))
#let parts = split-col(aug, 2)
#to-tyrst(parts.at(0)) | #to-tyrst(parts.at(1))
```

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mid \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Parameters

```
split-col(
  matrix: bytes content array int float str,
  index: int
) -> array
```

matrix bytes or content or array or int or float or str

Matrix to split.

index int

Zero-based first column of the right block.

5.5.17 primitive-part

Divide a rational-polynomial matrix by its content.

The result is the primitive matrix whose coefficient GCD has been removed.

```
#let x = symbol("x")
#let P = matrix(((mul(2, x), mul(4, x)),
(mul(6, x), mul(8, x))))
#to-tyrst(primitive-part(P))
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Parameters

```
primitive-part(matrix: bytes content array int float str) -> bytes
```

matrix bytes or content or array or int or float or str

Matrix whose entries are rational polynomials.

5.5.18 content

Compute the coefficient content of a rational-polynomial matrix.

The content is the common coefficient GCD and is returned as an atom payload.

```
#let x = symbol("x")
#let P = matrix(((mul(2, x), mul(4, x)),
(mul(6, x), mul(8, x))))
#to-typst(content(P))
```

2

 x

Parameters

`content`(`matrix`: bytes `content` array `int` `float` `str`) -> bytes

matrix bytes or `content` or array or `int` or `float` or `str`

Matrix whose entries are rational polynomials.

5.5.19 matrix-at

Read one matrix entry as an atom payload using zero-based indices.

```
#let A = matrix(((1, 2), (3, 4)))
#to-typst(matrix-at(A, 0, 1))
```

2

Parameters

```
matrix-at(
  matrix: bytes content array int float str,
  row: int,
  col: int
) -> bytes
```

matrix bytes or `content` or array or `int` or `float` or `str`

Matrix to index.

row `int`

Zero-based row index. Must be within the matrix bounds.

col `int`

Zero-based column index. Must be within the matrix bounds.

5.5.20 matrix-shape

Return the matrix shape as the two-element array (rows, columns).

```
#repr(matrix-shape(matrix(((1, 2), (3, 4)))))
```

(2, 2)

Parameters

`matrix-shape`(`matrix`: bytes `content` array `int` `float` `str`) -> array

matrix bytes or content or array or int or float or str

Matrix to inspect.

5.5.21 matrix-is-zero

Test exactly whether every matrix entry is zero.

This is an exact symbolic predicate, not a floating-point tolerance test.

```
#matrix-is-zero(matrix(((0, 0), (0, 0))))
```

true

Parameters

matrix-is-zero(matrix: bytes content array int float str) -> bool

matrix bytes or content or array or int or float or str

Matrix to inspect.

5.5.22 matrix-is-diagonal

Test exactly whether every off-diagonal matrix entry is zero.

Rectangular matrices are accepted; entries outside the main diagonal must be zero.

```
#matrix-is-diagonal(matrix(((1, 0), (0, 2))))
```

true

Parameters

matrix-is-diagonal(matrix: bytes content array int float str) -> bool

matrix bytes or content or array or int or float or str

Matrix to inspect.

5.5.23 matrix-derivative

Differentiate every matrix entry with respect to an indeterminate.

```
#let x = symbol("x")
#to-tyrst(matrix-derivative(matrix(((pow(x, 2),
x), (1, 0))), x))
```

$$\begin{pmatrix} 2x & 1 \\ 0 & 0 \end{pmatrix}$$

Parameters

```
matrix-derivative(
  matrix: bytes content array int float str,
  var: bytes content str
) -> bytes
```

matrix bytes or content or array or int or float or str

Rational-polynomial matrix to differentiate.

var bytes or **content** or **str**

Indeterminate with respect to which each entry is differentiated.

5.6 Scalar constructors

5.6.1 add

Construct an exact sum from expression values.

Arguments are converted as by `atom`. With no arguments, the result is the additive identity zero.

```
#to-tyrst(add("x", 1, "y"))
```

$$1 + x + y$$

Parameters

`add(..terms: arguments) -> bytes`

..terms arguments

Positional terms to add.

5.6.2 mul

Construct an exact product from expression values.

Arguments are converted as by `atom`. With no arguments, the result is the multiplicative identity one.

```
#to-tyrst(mul(2, "x", "y"))
```

$$2$$

$$x$$

$$y$$

Parameters

`mul(..factors: arguments) -> bytes`

..factors arguments

Positional factors to multiply.

5.6.3 neg

Negate an expression exactly.

```
#to-typst(neg("x"))
```

 -1
 x

Parameters

```
neg(expr: bytes content int float str) -> bytes
```

expr bytes or content or int or float or str

Expression to negate.

5.6.4 sub

Subtract rhs from lhs exactly.

```
#to-typst(sub("x", "y"))
```

 $x - y$

Parameters

```
sub(
  lhs: bytes content int float str,
  rhs: bytes content int float str
) -> bytes
```

lhs bytes or content or int or float or str

Minuend.

rhs bytes or content or int or float or str

Subtrahend.

5.6.5 div

Construct the exact quotient lhs / rhs.

```
#to-typst(div(1, "x"))
```

 x^{-1}

Parameters

```
div(
  lhs: bytes content int float str,
  rhs: bytes content int float str
) -> bytes
```

lhs bytes or content or int or float or str

Numerator.

rhs bytes or content or int or float or str

Denominator.

5.6.6 pow

Construct the exact power base^{exp} .

```
#to-typst(pow("x", 3))
```

 x^3

Parameters

```
pow(
  base: bytes content int float str,
  exp: bytes content int float str
) -> bytes
```

base bytes or content or int or float or str

Base expression.

exp bytes or content or int or float or str

Exponent expression.

5.7 Advanced configuration

5.7.1 init

Create an independent set of Tymbolica functions.

The returned dictionary exposes Tymbolica's parsing, algebra, evaluation, solving, and matrix operations. Use `init` when you want to select a symbol namespace, plugin location, or parser grammar; ordinary calculations can use the imported top-level functions directly.

```
#let sym = init(namespace: "physics")
#let symbol = sym.symbol
#let render = sym.canonical
#raw(render(symbol("x"), namespaces: true))
```

physics::x

Parameters

```
init(
  namespace: str,
  source: str bytes none,
  grammar: dictionary,
  notation: dictionary
) -> dictionary
```

namespace str

Default namespace for symbols parsed from Typst math or strings. A per-call namespace passed to `math` or `symbol` takes precedence.

Default: "typst"

source `str` or `bytes` or `none`

WebAssembly plugin path or bytes passed to Typst's plugin constructor. `none` selects the bundled engine. A custom source must be an uncompressed Wasm module; relative paths are resolved by Typst from this source file.

Default: `none`

grammar `dictionary`

Parser grammar used by `math` and `array-tree` unless they receive an explicit override.

Default: `_default_grammar`

notation `dictionary`

Default document-side notation used by `to-typst`. Build one with `notation`; a per-call override still takes precedence.

Default: `atom-render.notation()`

6 Compatibility and licensing

This manual describes Tymbolica 0.1.0. The package is tested with Typst 0.14 or newer.

Tymbolica's original source code is released under the [MIT License](#). Symbolica is developed by the Symbolica contributors and is distributed under its own [license terms](#). The MIT License does not relicense Symbolica or the bundled WebAssembly engines; Symbolica's terms still apply to their use.

For source, issues, and release history, visit github.com/lcnbr/tymbolica.

7 Acknowledgements

Tymbolica would not exist without [Symbolica](#). Thank you to its contributors for building and sharing the algebra engine at the heart of this package.

Thanks also to [Parsely](#), which makes it possible to work with mathematics written directly in Typst, and to [Tidy](#), which powers this manual's examples and reference pages.